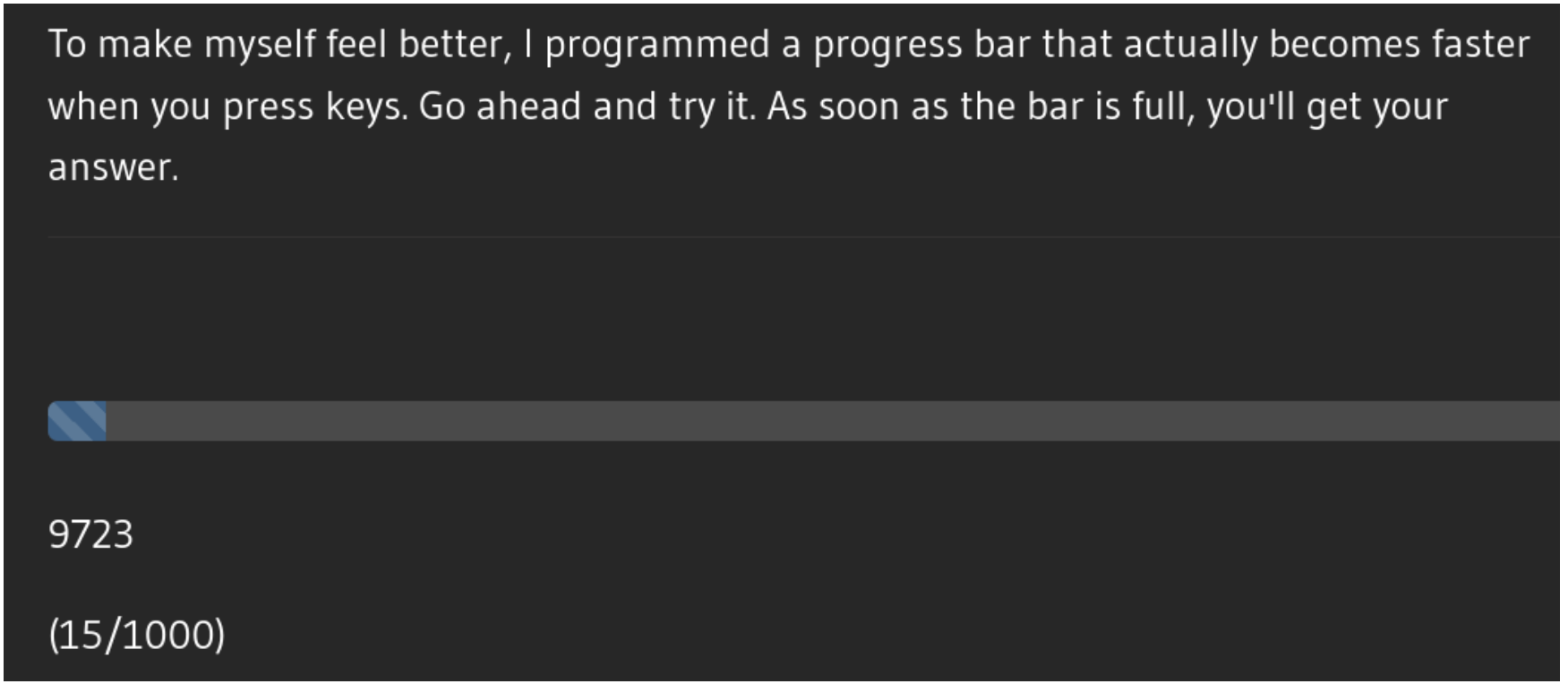
Hack the Web - ProgressBar



This challenge wants us to either wait 1000 seconds or press keys 1000 times to reveal the answer

Hack the Web – ProgressBar Inspecting the Code

```
let step = 1
let steps = 1000
let x = 3
```

```
function transform(x) {
  return (x * 11) % 10000
}
```

```
function work(noTimeout) {
  if (step >= steps) {
    bar.style.width = '100%'
  } else {
    step++
    bar.style.width = ((step / steps) * 98.9 + 1) + '%'
    x = transform(x)
    value.innerHTML = x.toString()
    status.innerHTML = '(' + step + '/' + steps + ')'
    if (!noTimeout) {
      setTimeout(work, 1000)
    }
  }
}
```

When we inspect the script that's being run on this page, we see that it's running a transformation on a number 999 times, once a second, then stops

Hack the Web – ProgressBar Inspecting the Code

```
window.onkeydown = (e) => {
  if (e.repeat) return
  work(true)
}
```

There's also a function in the script which advances the progress by 1 each time a key is pressed

Hack the Web – ProgressBar Simulating Keypresses /w JS

```
for (let i = 0; i < 990; i++)
  window.dispatchEvent(new
KeyboardEvent('keydown', {key: 'a',
  repeat: false}));
```

We can simulate keypresses with JavaScript code in the browser console, saving us a lot of time

Hack the Web – ProgressBar
Running the Progress Bar Function

```
while (step < steps) work(true);
```

We can also run the script's `work` function until all of the transformations have been completed

Hack the Web – ProgressBar Solving the Equation Directly

```
let step = 1
let steps = 1000
let x = 3
```

```
function transform(x) {
  return (x * 11) % 10000
}
```

```
function work(noTimeout) {
  if (step >= steps) {
    bar.style.width = '100%'
  } else {
    step++
    bar.style.width = ((step / steps) * 98.9 + 1) + '%'
    x = transform(x)
    value.innerHTML = x.toString()
    status.innerHTML = '(' + step + '/' + steps + ')'
    if (!noTimeout) {
      setTimeout(work, 1000)
    }
  }
}
```

This script essentially multiplies the number 3 by 11 by the number of steps (999), keeping it within the bounds of 10000 with a modulo operation

Hack the Web – ProgressBar Solving the Equation Directly

```
(3 * pow(11,999,10000)) % 10000
```

We can run the equation directly with a Python command and get the answer that way